

COMP433 Software Engineering · Phase 3 feedback

Used-Electronics Exchange Platform Group G | Section 6 | Business 62

Phase3-Feedback: G-G-s6

- [3.1] Actor Description: B · Nine actors all declared and used, but several rows list features rather than the role, and the email row narrates the platform. Check as a group.
- [3.2] Overall Use-Case Diagram: C · Tool named and boundary right, but two include ladders sequence a process, plus auth bubbles, an NFR-as-use-case and a misused generalisation. Check as a team.
- [3.3] Described use-cases: B- · Process Refund discriminates five failure modes well, but click triggers and field-check-as-error flows recur across the set. Check individually.
individual: Nadeen / Submit Repair Request: B · Marah / Process Refund: B+ · Anas / Purchase Device: B · Ramz / Submit Device for Sale-Exchange: C+ · Mariam / Create Appointment: C+
- [3.4] Overall Activity Diagram: C+ · The activity diagram was submitted as a separate file (Final Activity diagram.pdf), so it is assessed from there, not missing: six well-partitioned lanes with a System lane and guarded decisions, but the sale, repair and refund processes sit as three disconnected threads rather than one connected flow. Check as a team.

=> Contributors named per team task (3.1 Ramz; 3.2 Mariam; 3.4 Anas), each 3.3 spec carries its author.

Phase 2 uptake: The refund policy is now testable inside Process Refund, and the single 'manage appointments' need is split into Create, Modify and Cancel.

Overall (B-): actors and specs are sound and the activity model is broad; the include-decomposition use-case diagram and the disconnected activity flows are where it trails

The grades above are the team marks for this phase. The individual use-case work and the contribution notes are recorded, not separately marked yet; they are combined into each member's mark in the final report.

If the issues flagged on the team and individual tasks are addressed for the final report, those tasks may be reconsidered.



Faculty of Engineering and Technology

Computer Science Department

Software Engineering - COMP433

Semester 2 (Spring) 2025/2026

Phase 3: Requirements Analysis and Modelling

Used-Electronics Exchange Platform

Group members:

Name	Student ID	Section	Use case described (Task 3.3)
Anas Hamdan Al Haj	1230038	6	Purchase Device
Mariam Baradia	1230663	6	Create Appointment
Ramz Abufarha	1230963	6	Submit Device for Sale/Exchange
Nadeen Jaber	1220044	6	Submit Repair Request
Marah Taweel	1223094	6	Process Refund

Prepared by:

Group No: G | Section: 6

Business No: 62

Instructor: Dr Hisham Ihshaish | Submission date: 15/6/2026

Task 3.1: Actors and their descriptions

Lead : Ramz Abufarha

Contributors : [Anas Hamdan Al Haj: Reviewing], [Nadeen Jaber: Reviewing and discussing actor analysis], [Ramz Abufarha: Final formatting and integration] .

Actor	Semantic description
Customer	A registered user who interacts with the platform to browse available devices, purchase devices, submit devices for sale or exchange, schedule appointments, request repairs and deliveries, view device information, submit complaints, and track the status of submitted requests.
Sales Employee	An authorized employee responsible for managing device information within the system. The sales employee creates and updates device records, assists with device listing activities, and accesses business records according to assigned permissions.
Inspector	An authorized employee responsible for inspecting submitted devices and assessing their condition. The inspector records inspection results, assigns condition categories, identifies defects, and determines whether a device is eligible for sale or exchange.
Technician	An authorized employee responsible for handling repair services. The technician reviews repair requests, updates repair statuses, records repair progress, and manages repair completion or rejection information.
Delivery Employee	An authorized employee responsible for managing device deliveries. The delivery employee updates delivery status information and supports the delivery process for purchased or exchanged devices.
Accountant	An authorized employee responsible for financial operations within the platform. The accountant processes payments and refunds, maintains financial records, and supports transaction verification and auditing activities.
Manager	An authorized managerial user who oversees business operations. The manager accesses reports, reviews business records, monitors system activities, and supports operational decision-making.
Payment Gateway (External System)	An external payment processing system that authorizes, verifies, and confirms electronic payments associated with purchases, exchanges, repair services, and refund transactions.
Email Service (External System)	An external email service used by the platform to support password recovery and email-based notifications. It enables communication between the system and users through verified email addresses.

[3.1]
The Customer row reads as a feature list (browse, purchase, submit, schedule, request, view, complain, track) rather than one sentence on the role and how it interacts. Several rows below do the same. For the final report, describe each actor by its role and its relationship to the platform, and leave the individual goals to the use cases.

[3.1]
Payment Gateway is declared here and then wired into 3.2 and the Purchase and Refund specs, so it earns its row. The actor-completeness the calibration set kept missing!

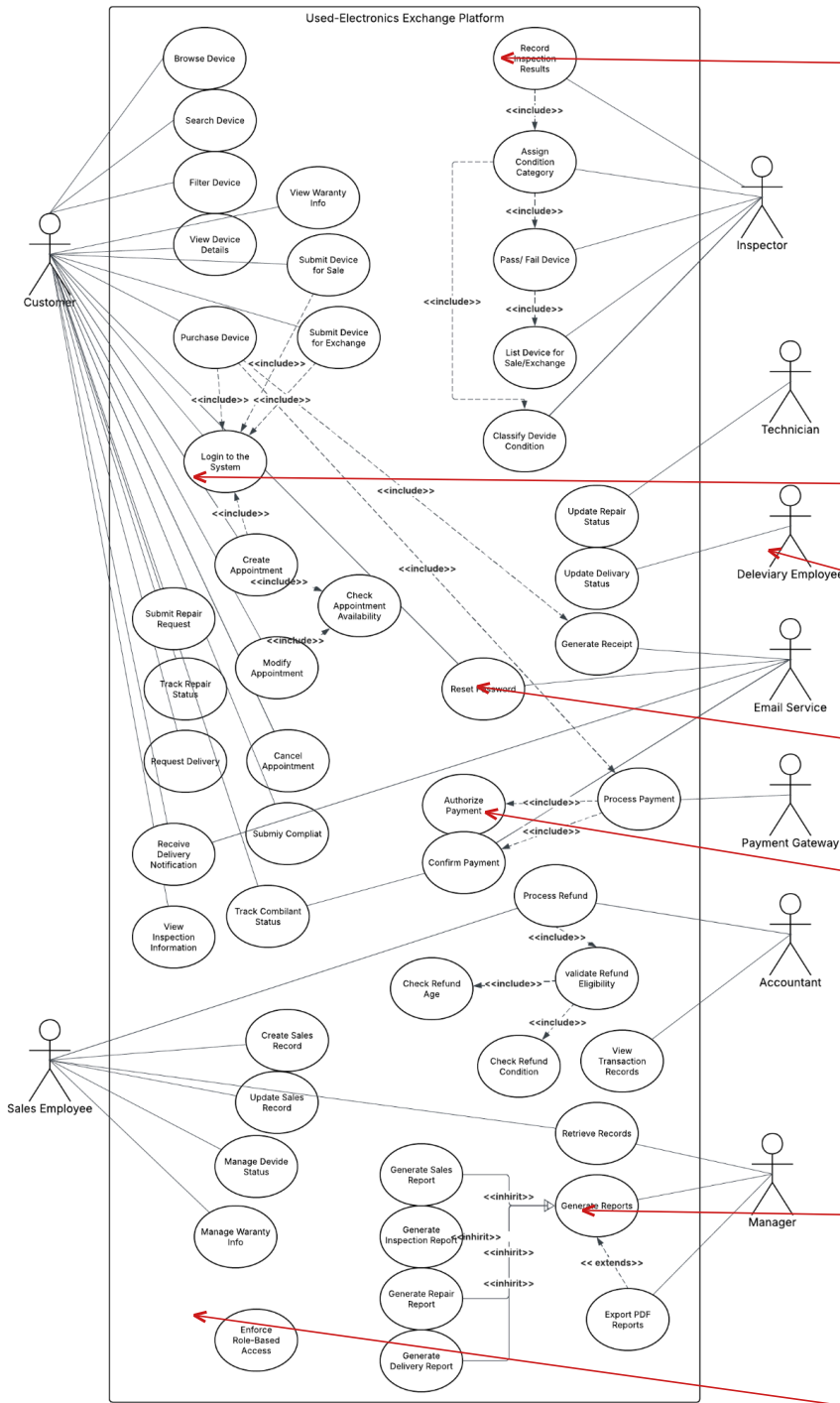
[3.1]
Email Service is described by what the platform does through it (recovery, notifications) rather than by its own role. Compare the Payment Gateway row, which leads with authorising payments. For the final report, reword it to the channel's own job: delivering verified messages to users.

Task 3.2: Overall system use case diagram

Lead: Mariam Baradeiya

Contributors: [Marah Taweel: Diagram organization and layout], [Ramz Abufarha: Requirements review, actor-to-use-case mapping, traceability verification, and use case validation]

UML Tool Used: Lucidchart (UML 2.x)



[3.2] Inspector branch (Record Inspection Results, Assign Condition Category, Pass/Fail Device, List Device, Classify Device): five cases strung with <<include>> to walk one actor's procedure top to bottom. Include is for shared sub-behaviour the base reuses, like Borrow including Compute Return-Dates, not for step order. For the final report, model this ordering in the activity diagram and leave only shared sub-cases as includes.

[3.2] Login to the System drawn as a use case and pulled into Purchase Device and Create Appointment by <<include>>. Authentication is plumbing with no business value to an actor, so it stays off the boundary. For the final report, make 'authenticated' a precondition and drop the bubble.

[3.2] The actor label 'Deleviary Employee' is misspelt on the boundary. For the final report, correct it to Delivery Employee.

[3.2] Reset Password drawn as a use case and wired to Email Service. Account recovery is not a goal the platform sets out to deliver, it is the same plumbing as Login. For the final report, take it off the boundary.

[3.2] Payment branch: Process Payment <<include>> Authorize Payment and Confirm Payment is the same fault, a sequence forced through include. Authorise-then-confirm is the flow of Process Payment, not two reusable cases. For the final report, collapse these into the Process Payment flow.

[3.2] Report cluster: Generate Sales/Inspection/Repair/Delivery Report each point to Generate Reports with <<inherit>> generalisation, and Export PDF Reports <<extends>> it. Generalisation says the children ARE kinds of the parent; here Generate Reports invokes each one, which is include. The labels also read 'inhirit' and 'extends'. For the final report, model the four as <<include>> of Generate Reports and fix the stereotype spelling.

[3.2] Enforce Role-Based Access is a system-wide constraint no actor invokes for value, so it is a non-functional requirement, not a use case. For the final report, record it as a constraint note or in your NFR section, not as an ellipse.

Task 3.3: Use case specifications (four or five key use cases)

Use case 1:

Field	Description
System Name	Used Electronics Exchange Platform
Use Case Title	Submit Repair Request
Author (group member)	Nadeen Jaber
Description	A customer may submit a repair request for a device that requires maintenance or repair. The customer provides device information and description of the problem. The system creates a repair request record, assigns a unique repair request identifier, and sets the request status to “Pending” for technician review.
Actors	Primary: Customer Secondary: Technician
Data	Repair request identifier; device information; problem description; submission date; customer identifier; repair request status.
Stimulus / Trigger	The customer selects "Submit Repair Request" from their account dashboard.
Pre-conditions	1. The customer is authenticated and logged into the system. 2. The customer is authorized to access repair request functions. 3. The system is able to access the device records and repair request storage, and can generate unique repair request identifiers.
Workflow / Sequence of Events	1. The customer selects "Submit Repair Request". 2. The system displays the repair request form. 3. The customer enters device information. 4. The customer enters a description of the problem. 5. The customer submits the repair request. 6. The system validates the entered information. 7. If valid: 7.1. The system generates a unique repair request identifier. 7.2. The system records the customer identifier and submission date. 7.3. The system sets the repair request status to "Pending".

[3.3]
Trigger written as a menu selection. The stimulus is the repair request being submitted for technician review, an event like a command issued on a borrower's behalf, not the menu selection; the click is step one of the flow. For the final report, phrase the trigger as that event.

	7.4. The system stores the repair request. 7.5. The system displays a confirmation message containing the repair request identifier. 8. Else (error path): 8.1. If required fields (device information or problem description) are missing, the system displays the message: "Please complete all required fields before submitting your repair request." 8.2. The repair request is not created. Alternative Flow A1 – Customer Cancels Submission A1.1 While completing the repair request form, the customer decides not to continue. A1.2 The customer selects the Cancel option. A1.3 The system asks the customer to confirm cancellation. A1.4 The customer confirms cancellation. A1.5 The system discards any unsaved information entered in the form. A1.6 The system returns the customer to the dashboard. A1.7 No repair request record is created. Exception Flow E1 – Device Already Has Active Repair Request E1.1 During validation, the system checks whether the specified device already has an active repair request or is currently under maintenance.
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[3.3] Exception E1 turns the one-active-repair-per-device rule into a pre-validation branch carrying the exact on-screen wording, the system-state check an error flow is for, not a field check. Well judged.

	E1.2 The system detects an existing active repair request or maintenance record. E1.3 The system prevents creation of a new repair request for the same device. E1.4 The system displays the message: "This device already has an active repair request or is currently under maintenance." E1.5 No repair request record is created. E1.6 The use case terminates. Exception Flow E2 – System Failure During Request Creation E2.1 After the customer submits the repair request, the system encounters an unexpected error while creating the repair request record. E2.2 The system logs the error for administrative review. E2.3 The system displays an error message indicating that the repair request could not be submitted at this time. E2.4 No repair request identifier is generated. E2.5 The repair request record is not created. E2.6 The use case terminates.
Post-conditions	Successful Completion • A repair request record exists in the system. • A unique repair request identifier has been assigned.

	• The repair request status is set to "Pending". • The repair request is available for technician review. Failed Completion • No repair request record is created. • No repair request identifier is assigned. • The customer remains on the repair request form or is returned to the dashboard. • Appropriate validation or error messages are displayed. • System errors are logged where applicable.
Comments	• This use case satisfies the repair request requirements defined in UR8 and the related detailed requirements in Phase 2. • Payment processing for repair services is handled separately through the payment process and is outside the scope of this use case. • After successful submission, authorized technicians may review the request and update its status to "In Progress", "Completed", or "Rejected" through separate maintenance processes. • Customers may monitor the status of submitted repair requests through their accounts.

Use case 2:

Field	Description
System Name	Used Electronics Exchange Platform
Use Case Title	Process Refund
Author (group member)	Marah taweel
Description	This use case describes how a registered customer requests a refund for a purchased device, a completed repair service, or a rejected exchange, and how the platform validates the request, routes it to the Accountant for review, processes the refund through the Payment Gateway, and updates the related transaction and financial records. It gives customers a way to get their money back when a purchase, exchange, or repair does not work out, while keeping the store's records accurate.
Actors	Customer (primary) Accountant Payment Gateway
Data	Refund record, the related transaction record, customer account, inspection/evidence report for defective items or rejected repairs, payment gateway transaction reference, notification log.
Stimulus / Trigger	The customer submits a refund request for an order, completed repair, or rejected exchange, because the item is defective, not as described, the repair was rejected, or the order was cancelled within the refund window. The use case can also be triggered automatically when an inspection finds that a returned device does not match the condition it was sold/exchanged in.
Pre-conditions	1.The customer is registered and logged in. 2.The order/repair/exchange transaction exists and is linked to a completed payment record. 3.The transaction is in a refund-eligible state (e.g., Delivered, Repair Completed, or Exchange Rejected after inspection). 4.The refund request is submitted within the refund-eligible window defined by the business refund policy. 5.No refund request is already open (Pending or Approved) for the same transaction.

[3.3]
Trigger spans the customer-initiated request and a system-detected mismatch on return, an event rather than a click, and the cleanest of the five. Keep it as the model for the others.

Workflow / Sequence of Events

Main Flow

1. The customer selects **"Request Refund"** for an eligible order, repair, or exchange transaction.
2. The system displays a refund request form containing the transaction details and the available refund reasons.
3. The customer selects a refund reason (defective device, repair rejected, item not as described, or cancelled order), optionally attaches supporting evidence, and submits the request.
4. The system validates the refund request against the refund eligibility rules, including:
 - Refund condition (transaction status and item condition).
 - Refund age (time elapsed since purchase/repair completion).
 - Absence of an existing open refund request for the same transaction.
5. If the request is eligible, the system creates a refund record with status **"Pending Review"** and records the submission timestamp.
6. The system notifies the Accountant that a new refund request requires review.
7. The Accountant reviews the refund request, transaction details, and any supporting evidence.
8. If the Accountant approves the refund request, the system determines the final refund amount according to the refund policy.
9. The system sends a refund instruction to the Payment Gateway, including the approved refund amount and the original transaction reference.
10. The Payment Gateway processes the refund transaction.
11. Upon successful confirmation from the Payment Gateway, the system:
 - Receives a refund reference number.
 - Updates the refund status to **"Completed"**.

- Updates the related transaction status to **"Refunded"**.
 - Updates the financial records accordingly.
12. The system sends a refund confirmation notification to the customer containing the refund amount and reference number.
13. The use case ends successfully.
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Alternative Flow A1 – Partial Refund Approved

A1.1 During review, the Accountant determines that the returned item is used, incomplete, damaged beyond normal use, or only partially satisfies the refund policy.

A1.2 The system calculates a reduced refund amount according to the refund policy.

A1.3 The Accountant approves the reduced refund amount.

A1.4 The system records the approved refund amount.

A1.5 The process continues from Step 9 of the Main Flow.

Alternative Flow A2 – Customer Cancels Refund Request

A2.1 While the refund request status is **"Pending Review"**, the customer chooses to withdraw the request.

A2.2 The customer selects **"Cancel Refund Request"**.

A2.3 The system updates the refund status to **"Cancelled by Customer"**.

A2.4 The system records the cancellation date and time.

A2.5 The use case ends.

Exception Flow E1 – Refund Request Not Eligible

E1.1 During validation, the system determines that the refund request does not satisfy one or more eligibility requirements.

E1.2 The system updates the refund status to **"Rejected – Not Eligible"**.

E1.3 The system records the reason for rejection.

E1.4 The system notifies the customer of the rejection and the corresponding reason.

E1.5 The use case ends.

Exception Flow E2 – Accountant Rejects Refund Request

E2.1 During review, the Accountant determines that the refund request does not satisfy the refund policy requirements.

E2.2 The Accountant rejects the refund request and provides a rejection reason.

E2.3 The system updates the refund status to **"Rejected"**.

E2.4 The system records the rejection reason.

E2.5 The system notifies the customer of the rejection and any available appeal options.

E2.6 The use case ends.

Exception Flow E3 – Payment Gateway Failure or Timeout

E3.1 The system sends the approved refund request to the Payment Gateway.

	E3.2 The Payment Gateway fails to process the refund or does not respond within the configured timeout period. E3.3 The system updates the refund status to "Failed – Retry Pending". E3.4 The system logs the failure details. E3.5 The system schedules an automatic retry. E3.6 If the maximum retry limit is reached, the system escalates the request to the Accountant for manual processing. E3.7 The customer is notified that the refund is delayed. E3.8 The use case ends pending manual resolution. Exception Flow E4 – System or Network Failure During Refund Processing E4.1 A system or network failure occurs before refund confirmation is received from the Payment Gateway. E4.2 The system rolls back any incomplete status updates. E4.3 The refund status remains "Pending Review". E4.4 The system records the failure in the system log. E4.5 The Accountant is notified for manual follow-up. E4.6 The use case ends pending resolution. Exception Flow E5 – Invalid Payment Reference
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[3.3]

Gateway timeout, network failure and an invalid reference each get a distinct error path and end state (Failed-Retry Pending, Pending Review, Requires Manual Refund). Distinguishing the failure modes rather than one catch-all is what lifts Process Refund clear of the rest!

	E5.1 The system cannot locate the original payment transaction reference or determines that the reference is invalid. E5.2 The system cannot process the refund automatically. E5.3 The system updates the refund status to "Requires Manual Refund". E5.4 The Accountant is notified to process the refund manually. E5.5 The customer is informed that additional processing time is required. E5.6 The use case ends pending manual resolution.
Post-conditions	1.Success: The refund record status is "Completed", the related transaction is updated to "Refunded" , the financial records reflect the refunded amount, the Payment Gateway reference is stored, and the customer has received a confirmation. 2.Failure (rejected/not eligible): The refund status is "Rejected" or "Rejected – Not Eligible", the transaction status is unchanged, and the customer has been notified with the reason. 3.Failure (payment issue): The refund status is "Failed – Retry Pending" or "Requires Manual Refund", the Accountant has been notified for follow-up, and the transaction stays "Pending Refund" until resolved.
Comments	This use case maps to UR12 ("The system shall support refunds according to the business refund policy") and its refinements SR12.1–SR12.4 from Phase 2 (refund processing by authorized employees, refund record fields, rejection of requests that fail policy, and updating the transaction record after a refund). The Phase 2 feedback noted that UR12/SR12 point at an unstated refund policy. To make this testable, we're proposing these as the observable refund conditions: the item is returned within 14 days of purchase/repair completion; the item passes a return inspection matching its condition

	at sale/handover; for repairs, the refund applies only if the repair wasn't completed, was rejected before work started, or the same fault recurs within the repair's warranty period. Assumes the refund policy (eligible reasons, time window, deduction percentages for used items) is configurable by a Manager and stored as system data. This use case uses <<include>> with "Validate Refund Eligibility," which itself includes "Check Refund Age" and "Check Refund Condition," matching the Task 3.2 diagram.
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Use case 3:

Field	Description
System Name	Used Electronics Exchange Platform
Use Case Title	Purchase Device
Author (group member)	Anas Hamdan Al Haj
Description	Allows an authenticated customer to purchase an available device, complete payment, and receive a purchase receipt.
Actors	Customer (Primary), Payment Gateway (Secondary), Delivery Employee
Data	Customer ID, Device ID, Transaction ID, Payment Information, Payment Amount, Transaction Date, Receipt Information, Delivery Information
Stimulus / Trigger	Customer selects the Purchase Device option for an available device.
Pre-conditions	1. Customer is authenticated. 2. Device exists in the system. 3. Device has passed inspection. 4. Device status is Available. 5. Payment service is operational.
Workflow / Sequence of Events	Main Flow 1. The customer browses the available devices listed in the system and selects a device of interest. 2. The system displays the device details, including device name, brand, model, condition category, selling price, warranty information, and current availability status. 3. The customer reviews the displayed information and selects the Purchase Device option. 4. The system verifies that the selected device is still available for purchase and has not been sold or reserved by another customer. 5. If the device is available, the system creates a pending transaction record and associates it with the authenticated customer.

[3.3]
Trigger is the selection again rather than the event (the purchase being initiated against an available device). The pre-conditions are proper system state: device exists, passed inspection, status Available. For the final report, bring the trigger to that same standard.

	6. The system displays the available payment methods to the customer. 7. The customer selects a payment method and enters the required payment information. 8. The system validates the entered payment information and submits the payment request to the Payment Gateway. 9. The Payment Gateway processes the payment request and returns the authorization result. 10. The system verifies that the payment has been successfully completed. 11. The system records the payment transaction, including the payment identifier, payment amount, payment method, transaction identifier, and payment date. 12. The system creates a purchase transaction record containing the transaction identifier, customer identifier, device identifier, payment amount, and transaction date. 13. The system updates the device status from Available to Sold. 14. The system updates the ownership records to associate the purchased device with the customer. 15. The system generates an electronic purchase receipt containing the transaction details and purchased device information. 16. The system displays a purchase confirmation message and provides access to the generated receipt. 17. The system sends a confirmation notification and receipt to the customer. 18. The use case ends successfully. Alternative Flow A1 – Customer Requests Delivery A1.1 During the purchase process, the customer selects the Request Delivery option. A1.2 The system displays a delivery information form. A1.3 The customer enters the delivery address, contact number, and preferred delivery date.
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	A1.4 The system validates the entered delivery information. A1.5 The system creates a delivery record associated with the purchase transaction. A1.6 The system assigns the delivery status Pending. A1.7 The purchase process continues from Step 6 of the Main Flow. Exception Flow E1 – Device No Longer Available E1.1 During the availability verification process, the system detects that the selected device is no longer available. E1.2 The system displays an error message informing the customer that the device has already been sold or reserved. E1.3 The system cancels the pending purchase process. E1.4 No transaction record is created. E1.5 The use case terminates. Exception Flow E2 – Payment Authorization Failure E2.1 The Payment Gateway rejects the payment request. E2.2 The system records the failed payment attempt for auditing purposes. E2.3 The system displays an error message indicating that payment authorization has failed. E2.4 The system allows the customer to retry using a different payment method or cancel the purchase. E2.5 The device remains available for purchase. E2.6 The use case terminates if the customer cancels the transaction.
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	Exception Flow E3 – Payment Service Unavailable E3.1 The system is unable to establish communication with the Payment Gateway. E3.2 The system records the communication failure in the system log. E3.3 The system displays a message indicating that the payment service is temporarily unavailable. E3.4 The purchase transaction is not confirmed. E3.5 The device status remains Available. E3.6 The customer is advised to retry the purchase later. E3.7 The use case terminates.
Post-conditions	Successful Completion: • A purchase transaction record exists. • A payment record exists. • The device status is updated to Sold. • Ownership records are updated. • An electronic receipt is generated. • A delivery record exists if delivery was requested. Failed Completion: • No completed purchase transaction exists. • The device remains available for purchase. • Failed payment or system errors are logged. • Ownership records remain unchanged.
Comments	This use case satisfies the purchasing and payment requirements defined in UR2, SR2.1–SR2.6, delivery requirement SR9.1, transaction recording requirements SR11.3–SR11.6, and notification requirement SR14.1.

Use case 4:

Field	Description
System Name	Used Electronics Exchange Platform
Use Case Title	Submit Device for Sale/Exchange
Author (group member)	Ramz Abufarha
Description	Allows an authenticated customer to submit a used device for sale or exchange. The system collects and validates the required device information, creates a submission record, assigns a unique submission identifier, and marks the device as Pending Inspection before inspection and condition assessment activities are performed.
Actors	Customer (Primary), Inspector (Secondary)
Data	Customer ID, Submission ID, Device Category, Device Brand, Device Model, Device Age, Condition Description, Device Images, Submission Type (Sale/Exchange), Submission Date, Submission Status
Stimulus / Trigger	The customer chooses to submit a used device for sale or exchange through the platform.
Pre-conditions	1. Customer is authenticated. 2. Customer account is active. 3. The system is operational. 4. Required device information is available. 5. At least one device image is available for upload.
Workflow / Sequence of Events	Main Flow 1. The customer selects " Submit Device for Sale/Exchange ". 2. The system displays the device submission form. 3. The customer selects whether the device is submitted for Sale or Exchange . 4. The customer enters the required device information, including category, brand, model, estimated age, and condition

	description. 5. The customer uploads one or more images of the device. 6. The customer submits the completed form. 7. The system validates the entered information. 8. The system verifies that all mandatory fields have been completed. 9. The system creates a device submission record. 10. The system assigns a unique submission identifier. 11. The system records the selected submission type (Sale or Exchange). 12. The system stores the submitted device information and uploaded images. 13. The system sets the submission status to "Pending Inspection". 14. The system places the submitted device in the inspection queue. 15. The system notifies the Inspector that a new device requires inspection. 16. The system displays a confirmation message containing the submission identifier. 17. The system places the device in the inspection queue. 18. The system notifies the Inspector that a new submission requires inspection. 19. The use case ends successfully. Alternative Flow A1 – Exchange Submission A1.1 Customer selects Exchange. A1.2 System requests preferred replacement device category. A1.3 Customer enters exchange preferences. A1.4 System records submission type as Exchange. A1.5 Continue with Step 4 of Main Flow.	[3.3] Steps 14-15 (queue, then notify Inspector) reappear almost verbatim at 17-18, so a reader cannot tell whether the device is queued and notified once or twice. For the final report, collapse this to a single queue-and-notify pair.
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	Alternative Flow A2 – Customer Cancels Submission A2.1 Before submitting the completed form, the customer decides not to continue. A2.2 The customer selects Cancel Submission. A2.3 The system asks for confirmation. A2.4 The customer confirms the cancellation. A2.5 The system discards any unsaved information. A2.6 The system returns the customer to the dashboard. A2.7 No submission record is created. Exception Flow E1 – Missing Required Information E1.1 The customer submits the form with one or more required fields missing. E1.2 The system highlights the missing fields. E1.3 The system displays validation messages describing the missing information. E1.4 The submission record is not created. E1.5 The customer remains on the submission form. Exception Flow E2 – No Device Image Uploaded E2.1 The customer submits the form without uploading a device image. E2.2 The system displays the message: "At least one device image is required." E2.3 The submission record is not created. E2.4 The customer remains on the submission form.
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[3.3]

E1 (missing fields) and E2 (no image) cast required-field validation as exception flows. An error flow is a system or business failure (a save fails, a rule refuses the request); a missing field is a normal-flow check behind a guard. For the final report, move both into the main flow's validation step and reserve the exception flows for the real system error in E3.

	Exception Flow E3 – System Error During Submission E3.1 The system encounters an unexpected error while processing the submission. E3.2 The system records the error in the system log. E3.3 The system displays an error message indicating that the submission could not be completed. E3.4 No submission identifier is assigned. E3.5 The submission record is not created. E3.6 The use case terminates.
Post-conditions	Successful Completion • A device submission record exists. • A unique submission identifier has been assigned. • The submission type (Sale or Exchange) is recorded. • The submitted device information is stored in the system. • The uploaded device images are stored. • The submission status is set to "Pending Inspection". • The device is placed in the inspection queue. • The Inspector is notified of the new submission. Failed Completion • No completed submission record exists. • No submission identifier is assigned.

	• The device is not submitted for inspection. • Uploaded information is not saved as a completed submission. • Validation errors or system errors are recorded where applicable. • Appropriate error messages are displayed to the customer.
Comments	• This use case satisfies UR3 and SR3.1–SR3.5. • Submission of a device does not guarantee acceptance for sale or exchange. • All submitted devices must undergo inspection and condition assessment before any transaction can proceed. • Devices that fail inspection are not eligible for sale or exchange in accordance with UR16. • Inspection activities are handled through separate inspection management processes. • The inspection outcome determines whether the submitted device may proceed to sale or exchange workflows.

Use case 5:

Field	Description
System Name	Used-Electronics Exchange Platform
Use Case Title	Create Appointment
Author (group member)	Mariam Baradeiya
Description	This use case allows an authenticated customer to create a new appointment for an inspection or exchange service at a selected branch location. The customer selects the appointment type, preferred date, time, and branch, and the system validates availability before confirming and recording the appointment.
Actors	Customer (primary); System (secondary)
Data	Read: customer account credentials, existing appointment records (to check conflicts), available branches and time slots. Written: new appointment record containing appointment identifier, appointment type, date, time, branch location, and customer identifier.
Stimulus / Trigger	The authenticated customer selects "Create Appointment" from the appointments menu.
Pre-conditions	1. The customer has a registered account in the system. 2. The customer is authenticated (logged in). 3. At least one branch location with available appointment slots exists in the system.
Workflow / Sequence of Events	Main Flow: 1. The customer navigates to the Appointments section and selects "Book New Appointment." 2. The system displays the appointment booking form. 3. The customer selects the appointment type (Inspection or Exchange). 4. The customer selects a branch location. 5. The system displays available dates and times for the selected branch. 6. The customer selects a preferred date and time slot. 7. The customer confirms the appointment details. 8. The system checks whether the selected date, time, and branch combination is already assigned to another confirmed appointment . 9. If no conflict is detected: 9.1. The system assigns a unique appointment identifier. 9.2. The system creates and saves the appointment record containing: appointment identifier, appointment type, date, time, branch location, and customer identifier. 9.3. The system displays a confirmation message with the

[3.3]
'System' cannot be a secondary actor: it is the thing under design, not an external party the use case reaches (a secondary actor is something like Payment Gateway). For the final report, drop it and let Create Appointment rest on Customer alone.

	appointment details. Alternative Flow — Scheduling Conflict Detected (Step 8): 8a.1. The system detects that the selected slot is already taken by another confirmed appointment. 8a.2. The system rejects the booking request and notifies the customer that the selected slot is unavailable. 8a.3. The system returns to Step 5 and prompts the customer to select a different date or time. Error Flow — Concurrent Booking (Step 8): 8b.1. Two customers attempt to book the same slot simultaneously. 8b.2. The system applies concurrency control to ensure only one booking succeeds. 8b.3. The customer whose request was not confirmed receives a notification that the slot is no longer available and is prompted to select another slot. Error Flow — Unauthenticated Access: If the customer is not logged in and attempts to create an appointment, the system redirects them to the login page before proceeding.
Post-conditions	conditionsSuccess: A new appointment record is stored in the system with all required fields. The selected time slot is marked as booked and is no longer available to other customers. The customer receives a booking confirmation. Failure (conflict / error): No appointment record is created. The time slot remains the same. The customer is informed of the issue and may retry with a different selection.
Comments	1. Appointment types are limited to those supported by the system (e.g., Inspection, Exchange); the form should not allow free-text entry for type. 2. The system must handle simultaneous booking attempts without creating duplicate records. 3. Only future appointments can be created; the system should prevent selection of past dates. 4. This use case covers creation only; modifying or cancelling an existing appointment is handled in separate use cases. 5. Navigation to complete this use case must not exceed five steps in total.

[3.3] Unauthenticated access is written as an error flow, but 'the customer must be logged in' is a pre-condition, a state required before the case starts, not a failure met partway through. You already list it under pre-conditions. For the final report, remove this branch and let the precondition carry it.

Task 3.4: Overall activity diagram

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The Platform contains a large number of use cases that support different business operations. Including every use case in a single activity diagram would make the model excessively complex, difficult to read, and harder to understand. Therefore, we focused on the five main use cases selected in Task 3.3 (Submit Device for Sale/Exchange, Manage Appointment, Purchase Device, Submit Repair Request, and Process Refund), as these represent the platform's core business processes and demonstrate the interaction between the main actors and system components. The remaining use cases are still part of the overall system and are represented in the use case diagram, but were excluded from the activity diagram to maintain clarity, readability, and proper diagram organization. Due to the size and level of detail of the activity diagram, a higher-quality and more readable version is included in the reply of the message(the other PDF file).

[3.4]
The activity diagram is in your separately-submitted Final Activity diagram.pdf, it did arrive, so it is graded (not zero): a broad six-lane model with a System lane and decisions, but the sale, repair and refund flows never connect into one process. For the final report, embed the diagram in the report itself and link the three flows through a single start and shared decisions.